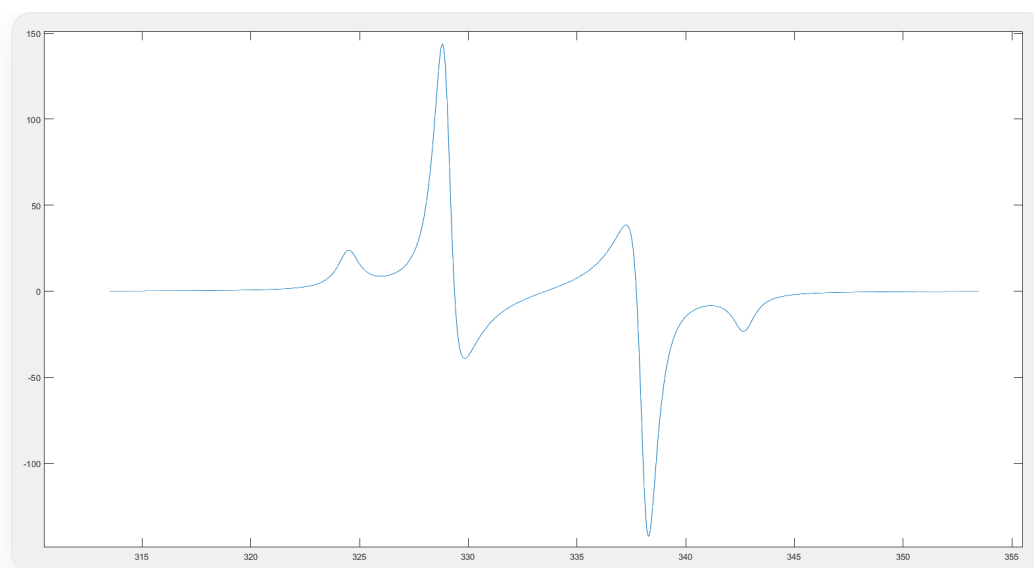


## Question

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A simulated EPR spectrum of a carbon-centered organic diradical in frozen toluene is shown below. Based on your analysis of the spectrum, determine the average distance (in pm) between the two single electrons of the diradical.



[This figure shows an EPR spectrum with magnetic field strength (in mT) on the x-axis and intensity (in arbitrary units) on the y-axis. The data points are as follows, with each point in a pair of parentheses, where the first number is the magnetic field strength (in mT) and the second number is the peak intensity (in arbitrary units):

|             |               |
|-------------|---------------|
| 313.1350000 | 1.4867028e-04 |
| 313.1353910 | 1.4972312e-04 |
| 313.1357820 | 1.5078641e-04 |
| 313.1361730 | 1.5186027e-04 |
| 313.1365640 | 1.5294486e-04 |
| 313.1369550 | 1.5404029e-04 |
| 313.1373460 | 1.5514673e-04 |
| 313.1377370 | 1.5626431e-04 |
| 313.1381281 | 1.5739318e-04 |
| 313.1385191 | 1.5853350e-04 |
| 313.1389101 | 1.5968540e-04 |
| 313.1393011 | 1.6084906e-04 |
| 313.1396921 | 1.6202461e-04 |
| 313.1400831 | 1.6321223e-04 |
| 313.1404741 | 1.6441208e-04 |
| 313.1408651 | 1.6562432e-04 |
| 313.1412561 | 1.6684913e-04 |
| 313.1416471 | 1.6808666e-04 |
| 313.1420381 | 1.6933710e-04 |
| 313.1424291 | 1.7060062e-04 |
| 313.1428201 | 1.7187740e-04 |
| 313.1432111 | 1.7316763e-04 |
| 313.1436022 | 1.7447149e-04 |
| 313.1439932 | 1.7578917e-04 |
| 313.1443842 | 1.7712087e-04 |
| 313.1447752 | 1.7846678e-04 |
| 313.1451662 | 1.7982709e-04 |
| 313.1455572 | 1.8120202e-04 |
| 313.1459482 | 1.8259177e-04 |
| 313.1463392 | 1.8399655e-04 |
| 313.1467302 | 1.8541657e-04 |
| 313.1471212 | 1.8685206e-04 |
| 313.1475122 | 1.8830323e-04 |
| 313.1479032 | 1.8977031e-04 |
| 313.1482942 | 1.9125353e-04 |
| 313.1486852 | 1.9275313e-04 |
| 313.1490762 | 1.9426934e-04 |
| 313.1494673 | 1.9580241e-04 |
| 313.1498583 | 1.9735258e-04 |
| 313.1502493 | 1.9892011e-04 |
| 313.1506403 | 2.0050525e-04 |
| 313.1510313 | 2.0210827e-04 |
| 313.1514223 | 2.0372943e-04 |
| 313.1518133 | 2.0536900e-04 |
| 313.1522043 | 2.0702726e-04 |
| 313.1525953 | 2.0870450e-04 |
| 313.1529863 | 2.1040099e-04 |
| 313.1533773 | 2.1211703e-04 |
| 313.1537683 | 2.1385292e-04 |
| 313.1541593 | 2.1560897e-04 |
| 313.1545503 | 2.1738548e-04 |
| 313.1549413 | 2.1918277e-04 |
| 313.1553324 | 2.2100117e-04 |
| 313.1557234 | 2.2284099e-04 |
| 313.1561144 | 2.2470258e-04 |
| 313.1565054 | 2.2658628e-04 |
| 313.1568964 | 2.2849244e-04 |
| 313.1572874 | 2.3042141e-04 |
| 313.1576784 | 2.3237355e-04 |
| 313.1580694 | 2.3434924e-04 |
| 313.1584604 | 2.3634885e-04 |
| 313.1588514 | 2.3837276e-04 |
| 313.1592424 | 2.4042138e-04 |
| 313.1596334 | 2.4249509e-04 |
| 313.1600244 | 2.4459432e-04 |
| 313.1604154 | 2.4671845e-04 |

2.4671948e-04)(3.1608065e+02 2.4887099e-04)(3.1611975e+02 2.5104929e-04)(3.1615885e+02  
 2.5325482e-04)(3.1619795e+02 2.5548804e-04)(3.1623705e+02 2.5774942e-04)(3.1627615e+02  
 2.6003942e-04)(3.1631525e+02 2.6235852e-04)(3.1635435e+02 2.6470723e-04)(3.1639345e+02  
 2.6708605e-04)(3.1643255e+02 2.6949549e-04)(3.1647165e+02 2.7193608e-04)(3.1651075e+02  
 2.7440835e-04)(3.1654985e+02 2.7691287e-04)(3.1658895e+02 2.7945018e-04)(3.1662805e+02  
 2.8202086e-04)(3.1666716e+02 2.8462551e-04)(3.1670626e+02 2.8726472e-04)(3.1674536e+02  
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 2.9817970e-04)(3.1690176e+02 3.0100122e-04)(3.1694086e+02 3.0386121e-04)(3.1697996e+02  
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 3.1570016e-04)(3.1713636e+02 3.1876338e-04)(3.1717546e+02 3.2186954e-04)(3.1721457e+02  
 3.2501946e-04)(3.1725367e+02 3.2821397e-04)(3.1729277e+02 3.3145391e-04)(3.1733187e+02  
 3.3474017e-04)(3.1737097e+02 3.3807362e-04)(3.1741007e+02 3.4145518e-04)(3.1744917e+02  
 3.4488580e-04)(3.1748827e+02 3.4836642e-04)(3.1752737e+02 3.5189805e-04)(3.1756647e+02  
 3.5548167e-04)(3.1760557e+02 3.5911834e-04)(3.1764467e+02 3.6280910e-04)(3.1768377e+02  
 3.6655506e-04)(3.1772287e+02 3.7035732e-04)(3.1776197e+02 3.7421703e-04)(3.1780108e+02  
 3.7813537e-04)(3.1784018e+02 3.8211355e-04)(3.1787928e+02 3.8615279e-04)(3.1791838e+02  
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 4.9345735e-04)(3.1877859e+02 4.9933528e-04)(3.1881769e+02 5.0531704e-04)(3.1885679e+02  
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 1.3006167e-01)(3.2863196e+02 1.3658361e-01)(3.2867107e+02 1.4250412e-01)(3.2871017e+02  
 1.4751554e-01)(3.2874927e+02 1.5127551e-01)(3.2878837e+02 1.5342689e-01)(3.2882747e+02  
 1.5362761e-01)(3.2886657e+02 1.5158854e-01)(3.2890567e+02 1.4711538e-01)(3.2894477e+02  
 1.4014686e-01)(3.2898387e+02 1.3078081e-01)(3.2902297e+02 1.1928007e-01)(3.2906207e+02

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-3.7848471e-02)(3.2968768e+02 -3.9539836e-02)(3.2972678e+02 -4.0702087e-02)(3.2976588e+02  
-4.1425017e-02)(3.2980499e+02 -4.1786391e-02)(3.2984409e+02 -4.1852765e-02)(3.2988319e+02  
-4.1680564e-02)(3.2992229e+02 -4.1317263e-02)(3.2996139e+02 -4.0802557e-02)(3.3000049e+02  
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-6.8948127e-04)(3.4740029e+02 -6.7994672e-04)(3.4743939e+02 -6.7060952e-04)(3.4747849e+02  
-6.6146409e-04)(3.4751760e+02 -6.5250504e-04)(3.4755670e+02 -6.4372716e-04)(3.4759580e+02  
-6.3512546e-04)(3.4763490e+02 -6.2669510e-04)(3.4767400e+02 -6.1843141e-04)(3.4771310e+02  
-6.1032990e-04)(3.4775220e+02 -6.0238622e-04)(3.4779130e+02 -5.9459617e-04)(3.4783040e+02  
-5.8695570e-04)(3.4786950e+02 -5.7946089e-04)(3.4790860e+02 -5.7210796e-04)(3.4794770e+02  
-5.6489324e-04)(3.4798680e+02 -5.5781320e-04)(3.4802590e+02 -5.5086442e-04)(3.4806500e+02  
-5.4404358e-04)(3.4810411e+02 -5.3734748e-04)(3.4814321e+02 -5.3077302e-04)(3.4818231e+02  
-5.2431719e-04)(3.4822141e+02 -5.1797708e-04)(3.4826051e+02 -5.1174988e-04)(3.4829961e+02  
-5.0563285e-04)(3.4833871e+02 -4.9962335e-04)(3.4837781e+02 -4.9371882e-04)(3.4841691e+02  
-4.8791677e-04)(3.4845601e+02 -4.8221479e-04)(3.4849511e+02 -4.7661054e-04)(3.4853421e+02  
-4.7110175e-04)(3.4857331e+02 -4.6568622e-04)(3.4861241e+02 -4.6036181e-04)(3.4865152e+02  
-4.5512646e-04)(3.4869062e+02 -4.4997813e-04)(3.4872972e+02 -4.4491489e-04)(3.4876882e+02  
-4.3993483e-04)(3.4880792e+02 -4.3503610e-04)(3.4884702e+02 -4.3021691e-04)(3.4888612e+02  
-4.2547552e-04)(3.4892522e+02 -4.2081023e-04)(3.4896432e+02 -4.1621940e-04)(3.4900342e+02  
-4.1170142e-04)(3.4904252e+02 -4.0725474e-04)(3.4908162e+02 -4.0287784e-04)(3.4912072e+02  
-3.9856925e-04)(3.4915982e+02 -3.9432753e-04)(3.4919892e+02 -3.9015129e-04)(3.4923803e+02  
-3.8603917e-04)(3.4927713e+02 -3.8198983e-04)(3.4931623e+02 -3.7800201e-04)(3.4935533e+02  
-3.7407443e-04)(3.4939443e+02 -3.7020589e-04)(3.4943353e+02 -3.6639518e-04)(3.4947263e+02  
-3.6264114e-04)(3.4951173e+02 -3.5894266e-04)(3.4955083e+02 -3.5529862e-04)(3.4958993e+02  
-3.5170795e-04)(3.4962903e+02 -3.4816960e-04)(3.4966813e+02 -3.4468256e-04)(3.4970723e+02  
-3.4124582e-04)(3.4974633e+02 -3.3785842e-04)(3.4978543e+02 -3.3451941e-04)(3.4982454e+02  
-3.3122786e-04)(3.4986364e+02 -3.2798287e-04)(3.4990274e+02 -3.2478357e-04)(3.4994184e+02  
-3.2162909e-04)(3.4998094e+02 -3.1851859e-04)(3.5002004e+02 -3.1545126e-04)(3.5005914e+02  
-3.1242629e-04)(3.5009824e+02 -3.0944291e-04)(3.5013734e+02 -3.0650036e-04)(3.5017644e+02  
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-2.8698508e-04)(3.5045015e+02 -2.8434369e-04)(3.5048925e+02 -2.8173700e-04)(3.5052835e+02  
-2.7916441e-04)(3.5056745e+02 -2.7662532e-04)(3.5060655e+02 -2.7411916e-04)(3.5064565e+02  
-2.7164535e-04)(3.5068475e+02 -2.6920335e-04)(3.5072385e+02 -2.6679261e-04)(3.5076295e+02

-2.6441260e-04)(3.5080205e+02 -2.6206280e-04)(3.5084115e+02 -2.5974270e-04)(3.5088025e+02  
 -2.5745180e-04)(3.5091935e+02 -2.5518962e-04)(3.5095846e+02 -2.5295569e-04)(3.5099756e+02  
 -2.5074952e-04)(3.5103666e+02 -2.4857068e-04)(3.5107576e+02 -2.4641870e-04)(3.5111486e+02  
 -2.4429315e-04)(3.5115396e+02 -2.4219360e-04)(3.5119306e+02 -2.4011964e-04)(3.5123216e+02  
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 -2.3207147e-04)(3.5138856e+02 -2.3011939e-04)(3.5142766e+02 -2.2819055e-04)(3.5146676e+02  
 -2.2628457e-04)(3.5150587e+02 -2.2440109e-04)(3.5154497e+02 -2.2253978e-04)(3.5158407e+02  
 -2.2070028e-04)(3.5162317e+02 -2.1888225e-04)(3.5166227e+02 -2.1708538e-04)(3.5170137e+02  
 -2.1530932e-04)(3.5174047e+02 -2.1355377e-04)(3.5177957e+02 -2.1181841e-04)(3.5181867e+02  
 -2.1010294e-04)(3.5185777e+02 -2.0840705e-04)(3.5189687e+02 -2.0673046e-04)(3.5193597e+02  
 -2.0507287e-04)(3.5197507e+02 -2.0343401e-04)(3.5201417e+02 -2.0181358e-04)(3.5205327e+02  
 -2.0021133e-04)(3.5209238e+02 -1.9862698e-04)(3.5213148e+02 -1.9706027e-04)(3.5217058e+02  
 -1.9551094e-04)(3.5220968e+02 -1.9397874e-04)(3.5224878e+02 -1.9246342e-04)(3.5228788e+02  
 -1.9096473e-04)(3.5232698e+02 -1.8948244e-04)(3.5236608e+02 -1.8801631e-04)(3.5240518e+02  
 -1.8656611e-04)(3.5244428e+02 -1.8513162e-04)(3.5248338e+02 -1.8371260e-04)(3.5252248e+02  
 -1.8230884e-04)(3.5256158e+02 -1.8092013e-04)(3.5260068e+02 -1.7954626e-04)(3.5263978e+02  
 -1.7818701e-04)(3.5267889e+02 -1.7684218e-04)(3.5271799e+02 -1.7551158e-04)(3.5275709e+02  
 -1.7419500e-04)(3.5279619e+02 -1.7289226e-04)(3.5283529e+02 -1.7160316e-04)(3.5287439e+02  
 -1.7032751e-04)(3.5291349e+02 -1.6906513e-04)(3.5295259e+02 -1.6781585e-04)(3.5299169e+02  
 -1.6657948e-04)(3.5303079e+02 -1.6535585e-04)(3.5306989e+02 -1.6414478e-04)(3.5310899e+02  
 -1.6294612e-04)(3.5314809e+02 -1.6175969e-04)(3.5318719e+02 -1.6058532e-04)(3.5322630e+02  
 -1.5942287e-04)(3.5326540e+02 -1.5827217e-04)(3.5330450e+02 -1.5713307e-04)(3.5334360e+02  
 -1.5600541e-04)(3.5338270e+02 -1.5488905e-04)(3.5342180e+02 -1.5378383e-04)(3.5346090e+02  
 -1.5268962e-04)(3.5350000e+02 -1.5160626e-04)

A. 174

B. 274

C. 374

D. 474

**E.** 574

**F.** 674

**G.** 774

**H.** 874

**I.** 974

**J.** 1074

**K.** 1174

**L.** 1274

## Answer

Correct Answer: **F**

## Detailed Explanation

The spectrum is a typical axially symmetric diradical frozen solution spectrum with zero-field splitting parameter  $E = 0$ .

### CHECKPOINT

1 PTS

Axially symmetric biradical with zero-field splitting parameter  $E = 0$

The average distance between spins  $\langle r \rangle$  can be estimated by the zero-field splitting parameter  $D$  value as  $D/mT = 2789/(\langle r/\text{Angstrom} \rangle^3)$ .

### CHECKPOINT

1 PTS

$$D/mT = 2789/(\langle r/\text{Angstrom} \rangle^3)$$

The peaks on both sides of the EPR spectrum correspond to z-direction absorptions, the distance between which is approximately equal to twice the  $D$  value. From the spectrum, we can read  $2D/mT \approx 18.18mT$ , so  $D = 9.09mT$ .

CHECKPOINT

1 PTS

$$D = 9.09mT$$

Substituting into the above relationship, we can obtain  $\langle r \rangle = 6.74 \text{Angstrom}$ , corresponding to 674 pm. Therefore, choose F.

CHECKPOINT

1 PTS

$$\langle r \rangle = 6.74 \text{Angstrom}$$